

Comprehensive Tutorial: Automating SAP Web Applications with Tricentis Tosca

This tutorial provides a detailed, step-by-step guide to automating **SAP Web Applications**—including SAP Fiori, SAP S/4HANA Web UI, and SAP Web GUI—using **Tricentis Tosca**. To maximize educational value, each section includes a theoretical background, definitions of professional terms, and practical examples.

1. Theoretical Foundation: Why Automate SAP Web Apps?

Before diving into the technical setup, it is essential to understand the context of SAP automation. Modern SAP environments are increasingly moving toward web-based interfaces like **SAP Fiori**. Unlike traditional desktop applications, web applications run in a browser and rely on HTML, CSS, and JavaScript.

Professional Terms:

- **SAP Fiori:** The modern user experience (UX) for SAP software, built on SAPUI5 technology, designed to be responsive and role-based.
- **SAP S/4HANA:** SAP's next-generation ERP suite that utilizes the Fiori interface for most user interactions.
- **Regression Testing:** A type of software testing that ensures that recent code changes have not adversely affected existing features.

Theoretical Background: Automation in SAP is critical because of the frequent updates and the complexity of business processes. Manual testing of a single "Order-to-Cash" cycle across multiple web screens is time-consuming and prone to human error. Automation ensures that these core business processes remain functional after every system update or patch.

2. Prerequisites and Environment Setup

To enable communication between the automation tool and the SAP system, specific configurations are required at the browser, server, and client levels.

2.1 Browser Connectivity

Term: Browser Extension – A small software module for customizing a web browser. Tosca uses the **Tricentis Automation Extension** to "see" and interact with the Document Object Model (DOM) of the webpage.

- **Action:** Install the extension for Chrome, Edge, or Firefox.
- **Example:** Without this extension, Tosca would only see the browser window as a single image rather than identifying individual buttons like "Save" or "Submit."

2.2 Server-Side Scripting (The "Handshake")

Term: RZ11 – A transaction code in SAP used to maintain profile parameters. **Term: Scripting** – The ability of an external program to control the SAP interface.

Theoretical Background: SAP has built-in security that prevents external tools from controlling its interface. Enabling "Scripting" is essentially giving Tosca permission to "drive" the application.

1. Open transaction **RZ11**.
2. Set `sapgui/user_scripting` to `TRUE`.
3. **Note:** This is typically done in Development or Quality environments, never in Production, to prevent accidental automated actions on live data.

3. The Model-Based Test Automation (MBTA) Approach

Tricentis Tosca is famous for its **Model-Based** approach, which differs significantly from **Script-Based** automation (like Selenium or UFT).

Component	Definition	Theoretical Background
Modules	Technical blueprints of a screen.	Instead of writing code to find a button, you "scan" the screen to create a Module. If the button moves, you update the Module once, and all tests are fixed.
Test Cases	The business logic sequence.	You build these by dragging Modules into a flow. This separates <i>how</i> to click

Component	Definition	Theoretical Background
		(Module) from <i>what</i> to do (Test Case).
TCPs	Test Configuration Parameters.	Variables that store environment data (e.g., URL, Browser type) so you can run the same test in different environments.

Example: If you have a "Login" Module, you can use it in 100 different Test Cases. If the "Login" button's ID changes, you only update the Module once, not 100 scripts.

4. Step-by-Step: Creating a Fiori Test Case

Step 1: Scanning with XScan

Term: XScan – The Tosca tool used to identify and capture UI elements from an application.

1. Open your Fiori Launchpad.
2. In Tosca, right-click a folder and select **Scan > Application**.
3. Select the browser window.
4. **Identification Strategy:** Fiori apps often have **Dynamic IDs** (IDs that change every time you refresh).
 - **Theory:** If an ID is `button_123` today and `button_456` tomorrow, the test will fail.
 - **Solution:** Use "Identify by Anchor" or "Identify by Index" in XScan to find the button based on its label or position instead of its volatile ID.

Step 2: Building the Logic

1. Create a new **TestCase**.
2. Drag the **OpenUrl** Standard Module to the start (to open the browser).
3. Drag your scanned Modules into the flow.
4. **ActionModes:**
 - **Input:** To type data into a field.
 - **Verify:** To check if a value is correct (e.g., checking if a "Success" message appears).
 - **Buffer:** To "copy" a value (like an Order Number) to use later in the test.

5. Advanced Techniques: Tables and Synchronization

5.1 Handling SAP Tables

Term: GridControl – A complex table structure used in SAP web apps. **Theory:** Tables are difficult because the data is dynamic. You might need to find a row where the "Material" is "Steel" and then click "Delete" in that specific row.

- **Example:** In Tosca, you use the syntax `$1` for the first row or `[Material==Steel]` to find a specific row dynamically.

5.2 Synchronization

Term: Steering – The process of Tosca interacting with the application. **Theory:** Web apps can be slow. If Tosca tries to click a button before the page finishes loading, the test fails. This is called a "Race Condition."

- **Solution:** Use the `WaitOn` ActionMode. This tells Tosca: "Don't move to the next step until this specific element is visible."

6. Best Practices for Professional Automation

1. **Modularization:** Don't scan a whole page. Scan the "Header," "Footer," and "Body" separately. This makes your Modules reusable across different apps.
2. **Data-Driven Testing:** Use **TestCase-Design (TCD)**.
 - **Theory:** Instead of creating 10 Test Cases for 10 different users, create 1 Test Case and link it to a data table. Tosca will run the test 10 times automatically.
3. **Naming Conventions:** Always name Modules after the SAP Transaction or Fiori App name (e.g., `VA01_CreateSalesOrder`).

7. Execution and Reporting

Term: Scratchbook – A temporary area to run and debug tests. **Term: ExecutionList** – A permanent log where formal test results are stored.

Theoretical Background: Testing isn't just about running the test; it's about the **Audit Trail**. In regulated industries (like Pharma or Finance), you must prove that the test passed.

- **ExecutionLists** provide a history of every run, including screenshots of failures, which is essential for compliance and debugging.

Final Note: For large-scale S/4HANA projects, use **Tricentis SAP Accelerators**. These are pre-built "Lego blocks" for standard SAP processes that allow you to start testing on Day 1 instead of building everything from scratch.

References: [1] Tricentis. (2026). *Tosca Commander User Manual*. <https://docs.tricentis.com> [2] SAP. (2025). *Fiori Design Guidelines*. <https://experience.sap.com/fiori-design/> [3] Merito. (2025). *SAP Automation Best Practices*. <https://www.merito.com/resources/blogs/how-to-start-sap-automation-with-tricentis-tosca-a-complete-guide>